The Aerial Guardian: Drone-Based Multi-Object Tracking for Aerial Surveillance

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Github: https://github.com/sanatwalia896/aerial_guardian_challenge

1. Problem Statement

Aerial surveillance systems face unique challenges compared to conventional fixed-camera monitoring. Pedestrians frequently occupy only a small number of pixels, camera motion introduces significant tracking instability, and real-time deployment imposes strict computational constraints.

The objective of this assignment was to design a robust drone-based multi-object tracking system capable of detecting and tracking pedestrians in VisDrone sequences while addressing:

- Small object detection
- Camera ego-motion
- Occlusions and identity switches
- Deployment constraints (<300 MB model budget)

2. Proposed Approach

The final system combines detection, tracking, motion compensation, and deployment optimization into a unified pipeline.

Pipeline

Drone Frame
↓
Adaptive SAHI
↓
YOLO11s + P2 Head
↓
ORB + RANSAC Ego-Motion Compensation
↓
Kalman Prediction
↓
Hungarian Association
↓
Lost Track Recovery
↓
OSNet ReID (Experimental)
↓
Tracked Output

Detector

A YOLO11s detector was fine-tuned on the VisDrone dataset for pedestrian detection.

Key modifications:

- Fine-tuning on aerial imagery
- P2 detection head for improved small-object detection
- Adaptive SAHI for distant pedestrians

These modifications improve recall for pedestrians that occupy only a small number of pixels.

Tracking

The tracking component follows a BoT-SORT-inspired design consisting of:

- Kalman motion prediction
- Hungarian assignment
- Lost-track recovery
- Track lifecycle management

The tracker is designed to maintain stable identities under drone-induced camera motion and temporary occlusions.

Ego-Motion Compensation

Traditional tracking systems often assume a stationary camera.

To address drone movement, ORB feature matching and RANSAC homography estimation were integrated into the pipeline.

Benefits include:

- Rotation handling
- Scale handling
- Perspective correction
- Reduced tracking drift

Appearance Modeling

An OSNet-based ReID module was integrated as an experimental appearance encoder.

The generated embeddings can be used to improve identity consistency in challenging situations such as:

- Pedestrian crossings
- Occlusions
- Dense crowds

3. Training Configuration

Parameter	Value
Base Model	YOLO11s
Dataset	VisDrone DET
Target Class	Person
Epochs	50
Image Size	640
Batch Size	16
Framework	Ultralytics

The detector was trained using VisDrone pedestrian data to improve adaptation to aerial imagery.

4. Results

Detection

The fine-tuned detector demonstrated significant improvement in aerial pedestrian detection compared to a generic COCO-pretrained model.

The inclusion of:

- P2 head
- VisDrone fine-tuning
- Adaptive SAHI

improved small-object recall and localization quality.

Tracking

The proposed tracker successfully maintained identities across challenging aerial sequences while handling:

- Camera motion
- Temporary occlusions
- Dense pedestrian scenes

Trajectory visualization and track persistence demonstrated stable tracking behavior throughout the evaluated sequences.

Small Object Handling

Aerial pedestrians often occupy only 15–25 pixels.

Adaptive SAHI was introduced to selectively enable sliced inference when required, improving distant pedestrian detection while minimizing unnecessary computational overhead.

5. Failure Analysis

Despite strong overall performance, several challenging cases remain:

Dense Crowds

Close pedestrian interactions may lead to identity ambiguity and occasional ID switches.

Long Occlusions

Extended disappearance of a target can result in track fragmentation.

Tiny Distant Pedestrians

Extremely small targets remain difficult to detect reliably.

Motion Blur

Rapid drone movement can reduce image quality and detector confidence.

These observations provide opportunities for future improvements.

6. Deployment Readiness

To support edge deployment, both major network components were exported to ONNX format:

- YOLO11s Detector → ONNX
- OSNet ReID → ONNX

Potential deployment targets include:

- NVIDIA Jetson Nano
- NVIDIA Xavier NX
- NVIDIA Orin NX

The complete system remains within the specified deployment budget and is suitable for further TensorRT optimization.

7. Engineering Tradeoffs

Several engineering decisions were made to balance accuracy and efficiency:

Choice	Benefit	Cost
YOLO11s	Lightweight deployment	Lower accuracy than larger models

Adaptive SAHI	Better small-object recall	Increased inference time
ORB + RANSAC	Improved ego-motion handling	Additional CPU overhead
ReID Integration	Better identity consistency	Higher computational cost
ONNX Export	Deployment flexibility	Export and validation complexity

8. Conclusion

The Aerial Guardian demonstrates a complete drone-based pedestrian detection and tracking pipeline tailored for aerial surveillance.

The final system combines:

- Fine-tuned YOLO11s detection
- Small-object optimization
- Ego-motion compensation
- BoT-SORT-inspired tracking
- Experimental appearance modeling
- Edge deployment support

The resulting solution provides a practical balance between tracking robustness, deployment efficiency, and aerial-scene adaptability while remaining suitable for future real-time deployment on embedded platforms.